

TAIN RANGE, United Kingdom

WMO: 030620

Lat: 57.8189N

Lon: 3.9667W

Elev: 13

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	25.8	20.4	15.3	24.6	23.4	17.7	29.1	38.2	43.3	32.8	42.3	3.4	250	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	70.0	61.6	67.4	59.8	65.2	58.6	63.4	68.0	61.6	65.6	59.9	63.7	12.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	81.8	65.5	59.7	76.5	63.6	58.1	72.2	61.9	28.8	68.0	27.5	65.7	26.4	63.8	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.7	25.0	22.1	17.2	76.9	5.4	2.9	13.3	79.0	10.1	80.7	7.0	82.3	3.1	84.4	
			DB												
			WB	16.9	68.6	5.5	4.9	12.9	72.1	9.7	75.0	6.6	77.7	2.5	81.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.1	38.3	39.1	41.8	45.2	49.0	54.0	57.4	57.2	53.6	48.2	42.4	38.1
	DBStd	8.52	5.40	5.36	4.90	4.87	4.56	4.14	3.69	3.81	4.10	4.97	5.54	6.76
	HDD50	1880	363	307	258	158	73	11	0	1	13	94	232	370
	HDD65	6545	828	725	720	594	496	330	236	244	341	520	677	834
	CDD50	813	0	2	3	14	42	131	231	224	122	38	5	1
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	18	0	0	0	0	0	2	9	7	0	0	0	0
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	10.5	11.2	11.8	11.9	11.2	10.4	10.1	9.0	9.0	9.7	10.5	10.5	10.5
Precipitation	PrecAvg	30.9	3.0	2.4	2.2	2.2	2.4	2.1	2.4	2.5	2.2	3.3	2.9	2.9
	PrecMax	35.5	6.0	4.8	4.4	3.9	4.8	4.3	4.6	5.3	3.8	7.8	4.3	4.5
	PrecMin	25.0	0.7	0.9	0.4	0.9	0.7	0.8	0.9	0.8	0.9	1.1	1.3	0.8
	PrecStd	3.0	1.4	0.9	0.9	0.8	1.1	0.9	1.1	1.0	0.9	1.6	0.8	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	56.1	58.3	63.0	67.4	71.2	74.7	74.0	69.6	63.7	58.2	55.4
		MCWB	49.4	50.9	50.6	54.9	57.8	60.3	65.1	65.5	61.7	57.4	53.6	51.6
	2%	DB	51.0	53.6	54.5	59.1	63.5	66.8	70.1	69.4	65.7	60.5	55.2	52.7
		MCWB	47.7	49.4	48.6	52.2	55.8	58.3	62.3	61.8	58.6	55.4	51.3	49.3
	5%	DB	48.7	50.6	52.2	56.3	60.1	64.1	67.1	66.6	63.2	58.2	53.0	50.5
		MCWB	45.7	46.9	47.3	50.0	53.7	57.0	60.4	59.8	57.3	53.8	49.6	47.2
	10%	DB	46.3	47.7	50.0	53.8	57.4	61.6	64.7	64.3	61.1	56.2	51.0	48.1
		MCWB	43.7	44.2	45.5	48.5	51.8	55.5	59.0	58.4	55.9	52.4	48.1	45.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	51.9	52.3	56.9	60.3	62.7	67.0	66.8	63.0	58.7	54.3	52.0
		MCDB	52.7	55.3	56.9	61.0	64.8	67.6	71.3	71.0	67.1	62.0	57.5	54.7
	2%	WB	48.1	49.6	49.5	53.1	56.8	59.8	63.8	63.0	60.3	56.5	52.0	49.7
		MCDB	50.6	53.1	53.6	57.6	61.8	64.7	68.3	67.8	64.2	59.5	54.6	52.4
	5%	WB	45.9	47.2	47.7	50.9	54.6	58.0	61.8	61.1	58.4	54.7	50.0	47.5
		MCDB	48.4	50.5	51.5	55.4	59.0	62.6	65.8	64.9	62.0	57.5	52.3	50.2
	10%	WB	44.0	44.5	45.9	49.0	52.6	56.5	60.0	59.5	56.8	53.0	48.4	45.4
		MCDB	46.3	47.4	49.5	53.1	56.4	60.6	63.6	63.1	60.0	55.6	50.7	48.0
Mean Daily Temperature Range	5% DB	MDBR	9.4	11.2	12.5	13.6	14.2	12.9	13.0	13.6	13.5	12.5	10.8	10.2
		MCDBR	10.6	13.1	15.6	17.4	18.7	17.3	17.8	16.9	16.5	14.2	12.7	12.1
		MCWBR	9.1	10.4	11.2	11.6	12.1	10.7	11.2	10.6	11.2	10.9	10.7	10.6
	5% WB	MCDBR	10.5	12.6	14.5	16.6	17.5	15.9	16.2	15.1	15.0	13.6	12.3	12.2
		MCWBR	9.2	10.4	11.0	11.8	12.4	10.9	11.5	10.6	11.5	11.4	10.9	10.8
Clear-Sky Solar Irradiance	taub	0.314	0.337	0.359	0.379	0.379	0.375	0.378	0.366	0.353	0.338	0.315	0.301	
	taud	2.215	2.306	2.314	2.303	2.352	2.396	2.398	2.439	2.473	2.422	2.269	2.152	
	Ebn at Noon	165	220	249	263	271	274	270	266	252	219	164	129	
	Edh at Noon	16	23	30	35	36	35	34	30	25	20	15	13	
All-Sky Solar Radiation	RadAvg	123	318	647	1087	1469	1440	1371	1091	753	394	163	84	
	RadStd	16	35	60	90	94	126	116	69	62	36	14	10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page